

Experimental data for a liquid/vapor binary mixture of pentane (P) and heptane (H) is given in the table. The quantity z_{liq} gives the mole fraction of pentane in the liquid phase while z_{vap} gives the mole fraction of pentane in the vapor phase. The data can be linearized using the relation $Y = A X$ where A is the separation factor, $X = z_{liq}/(1 - z_{liq})$, and $Y = z_{vap}/(1 - z_{vap})$.

For a general straight line with the intercept equal to zero $y = a x$, the "best fit" slope for perpendicular offsets is given by $a = -\bar{B} \pm \sqrt{\bar{B}^2 + 1}$ where $\bar{B} = \frac{S_{xx} - S_{yy}}{2S_{xy}}$ and the

averages are defined as $S_{xx} = \frac{1}{n} \sum_{i=1}^n x_i^2$, $S_{yy} = \frac{1}{n} \sum_{i=1}^n y_i^2$, and $S_{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i$, respectively. If $S_{xx} - S_{yy} > 0$, use the plus sign in front of the square root in the expression for the slope. If $S_{xx} - S_{yy} < 0$, use the minus sign.

Mole Fraction z_{liq}	Mole Fraction z_{vap}
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271

- Give the units on the mole fractions z_{liq} and z_{vap} .
- Give the units on the mole fraction ratios X and Y .
- Calculate the "best fit" value of the separation factor using perpendicular offsets as described above. Show all of your work.

(a)

unit of

$$z_{liq} = \frac{\text{mol P liq}}{\text{mol mix liq}}$$

$$z_{vap} = \frac{\text{mol P vap}}{\text{mol mix vap}}$$

(b)

$$\frac{X}{Y} = \frac{z_{liq}}{1 - z_{liq}} \cdot \frac{1 - z_{vap}}{z_{vap}}$$

$$\text{units: } \frac{\text{mol P liq}}{\text{mol mix liq} - \text{mol P liq}} \cdot \left(\frac{\text{mol mix vap} - \text{mol P vap}}{\text{mol P vap}} \right)$$

(c)

X	Y
6.52	61.5
1.46	12.3
0.661	5.10
0.340	2.34
0.170	1.00
0.0627	0.371

$$S_{xx} = \frac{1}{6} \sum_{i=1}^6 X^2 = 7.74$$

$$S_{yy} = 66.1$$

$$S_{xy} = \frac{1}{6} \sum_{i=1}^6 X Y = 70.95$$

$$B = \frac{S_{xx} - S_{yy}}{2S_{xy}} = -9.21$$

$$A = -B - \sqrt{B^2 + 1} = 9.21 - \sqrt{9.21^2 + 1} = -18.5$$

